APPLICATION OF QUANTUM COMPUTING IN DIGITAL EDUCATION SYSTEMS

Introduction

The rapid growth of digital education systems has transformed the way students' access, consume, and interact with educational content. Modern learning platforms continuously generate large volumes of student-related data, including academic performance, learning progress, course completion history, interests, preferences, strengths, weaknesses, and interaction patterns. The availability of such data creates significant opportunities for personalized learning and intelligent educational decision-making. However, effectively processing this information and identifying the most suitable learning resources or learning paths for individual students remains a challenging computational problem, particularly when the number of students, courses, learning resources, and educational constraints increases.

Personalized learning requires more than simply recommending individual courses or resources. An effective learning path must consider multiple factors simultaneously, including prerequisite relationships, student performance, learning objectives, interests, course difficulty, and expected learning outcomes. As the number of available educational resources increases, the number of possible combinations and sequences can grow rapidly, resulting in a complex optimization problem. Conventional approaches, including rule-based systems, classical machine learning, greedy algorithms, genetic algorithms, collaborative filtering, and integer programming, have been widely explored for educational recommendation and personalization. Although these approaches can provide effective solutions for many practical scenarios, their computational requirements may increase significantly as the problem size and number of constraints grow.

Quantum Computing (QC) has emerged as a potential computational paradigm for addressing certain classes of complex optimization problems. Unlike conventional computing, quantum computing uses quantum-mechanical principles such as superposition, entanglement, and interference to process information. Quantum optimization algorithms are therefore being investigated for problems involving large and complex solution spaces. However, current quantum hardware remains constrained by factors such as limited qubit availability, noise, circuit depth, and connectivity. Consequently, a purely quantum solution is not necessarily practical for large-scale digital education systems at present.

To address these limitations, this study proposes a **hybrid classical-quantum framework for digital education systems**, in which classical and quantum computing components perform complementary tasks. The classical component is responsible for student-data management, preprocessing, prerequisite verification, constraint handling, and validation of candidate solutions. The quantum component is utilized specifically for computationally intensive optimization tasks. In particular, the **Quantum Approximate Optimization Algorithm (QAOA)** is considered for optimizing personalized learning paths. The learning-path selection

problem can be formulated as a **Quadratic Unconstrained Binary Optimization (QUBO)** model, where binary variables represent educational decisions and the objective function incorporates factors related to learning suitability and educational constraints.

Within the proposed framework, the QUBO formulation is provided to the QAOA-based optimization process to generate high-quality candidate solutions. These solutions are subsequently evaluated by the classical component to verify prerequisites, constraints, and expected learning outcomes. This division of responsibilities allows the system to exploit quantum optimization capabilities while retaining the flexibility and reliability of classical computing for data processing and educational decision-making. The resulting system aims to generate personalized and optimized learning paths rather than relying solely on independent resource recommendations.

The proposed approach is intended to investigate the potential of quantum optimization in addressing increasingly complex personalization problems in digital education. In particular, the study considers the possibility of improving learning-path solution quality for sufficiently complex optimization instances. An improvement range of approximately 15–25% is considered as an expected potential outcome; however, such improvement is dependent on factors including problem size, QUBO formulation, algorithm parameters, and the capabilities of available quantum hardware and therefore requires experimental validation against appropriate classical baselines.

The remainder of this paper presents the related work and existing approaches to personalized digital education, followed by the proposed hybrid classical-quantum architecture and QUBO formulation. The QAOA-based optimization process and evaluation methodology are then discussed, followed by the expected results, limitations, and potential future directions for applying quantum computing to intelligent digital education systems.

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